

Nyayamitra: AI-Powered Legal Assistant for Case Analysis, RAG-Based Case Retrieval, and Fake News Detection

1st Nithish Kumar V, Member IEEE
IIITDM, Kancheepuram
Chennai, India
0009-0001-7882-921X

2nd Telaprolu Lakshmi Srinivas
IIITDM, Kancheepuram
Chennai, India
lakshmisrinivas365@gmail.com

3rd Sivaselvan B
IIITDM, Kancheepuram
Chennai, India
0000-0002-6333-3238

Abstract—This research presents NyayaMitra, an AI-powered legal assistant designed to improve the efficiency, precision, and accessibility of legal research within the Indian judicial context. Addressing the challenges posed by unstructured legal texts, the lack of structured legal databases, and the proliferation of misinformation, NyayaMitra integrates multiple AI capabilities including a legal chatbot, Retrieval Augmented Generation (RAG)-based precedent search, legal document summarization, and fake legal news detection. The system leverages the Phi-3 Mini model, fine-tuned on more than 200,000 Indian court judgments, to provide domain-specific accuracy. The RAG based retrieval system incorporates legal domain knowledge to improve relevance in case law search. Legal document summarization and misinformation detection modules were trained using supervised and prompt-based fine-tuning, respectively, achieving high factual accuracy and classification performance. The system is deployed using a scalable cloud infrastructure via Google Cloud, FastAPI backend, and cross-platform applications for web and mobile access. Evaluation results show significant performance gains over traditional legal research methods and general-purpose AI models, with up to 70% time savings reported by users. NyayaMitra establishes a robust foundation for domain-specific AI deployment in Indian legal research and paves the way for future multilingual, agentic, and scalable enhancements.

Index Terms—AI in Legal Research, Retrieval-Augmented Generation, Legal Document Summarization, Fake Legal News Detection, Indian Judiciary, Legal NLP, Domain-Specific Language Models, Legal Chatbot

I. INTRODUCTION

India's judiciary struggles with over 50 million pending cases, compounded by inefficient legal research and limited access to structured databases. Traditional keyword-based search engines fail to capture the nuanced semantics of legal queries, often producing incomplete results. Additionally, fake legal news through informal channels complicates legal discourse. This research introduces NyayaMitra, an AI-powered legal assistant designed for the Indian legal ecosystem. The system provides four core functionalities: a legal chatbot for statutory queries, RAG-based precedent search, document summarization, and fake news detection. NyayaMitra's key innovation is fine-tuning the lightweight Phi-3 Mini model on a curated Indian legal corpus, ensuring both contextual relevance and CPU-based efficiency for mobile platforms. The system

utilizes a comprehensive dataset of over 200,000 Indian court judgments from the Supreme Court and High Courts. The enhanced RAG architecture optimizes query processing through semantic rewriting, vector-based search using BAAI/bge-large-en embeddings, and context-aware response generation. The misinformation detection component employs prompt-based fine-tuning on authentic and fake legal claims from social media. This paper presents the dataset construction, system design, and evaluation results, demonstrating significant improvements over traditional legal research methods and establishing a foundation for AI-assisted legal technologies in India.

II. RELATED WORK

The integration of AI and NLP technologies into legal systems has seen rapid progress in addressing challenges related to legal research, document analysis, and misinformation detection. This section surveys foundational and recent work in four key domains relevant to this research.

A. AI in Legal Research and Case Analysis

Traditional legal research is primarily reliant on manual, keyword-based searches, which often lack semantic understanding and contextual relevance. Domain-specific legal NLP models, such as CaseLawBERT and ECHR-BERT, have demonstrated superior performance over general-purpose models by leveraging legal-domain corpora. However, these models are predominantly trained on Western legal systems, limiting their applicability to jurisdictions like India.

Chalkidis et al. [1] showed that training models on country-specific legal datasets significantly improves their performance in legal tasks. Similarly, Zhong et al. [2] emphasized the consistent outperformance of fine-tuned models over general-purpose ones in tasks such as judgment prediction and precedent retrieval.

In the Indian legal domain, Malik et al. [3] highlighted the urgent need for AI tools tailored to India's unique legal structure and data. Addressing this gap, NyayaMitra fine-tunes the Phi-3 Mini model on Indian legal texts, enabling

more accurate legal case retrieval and nuanced contextual understanding.

B. Retrieval-Augmented Generation (RAG) for Case Precedent Search

Conventional legal databases often rely on Boolean or keyword-based searches, which can yield irrelevant or incomplete results. Retrieval-Augmented Generation (RAG) mitigates this by integrating dense document retrieval with generative language modeling combining precision with semantic understanding.

Lewis et al. [4] introduced the RAG framework, demonstrating its ability to generate more accurate and grounded responses by retrieving supporting evidence from a large corpus. Wang et al. [5] further optimized this approach by exploring best practices in dense retrieval, achieving notable improvements in retrieval efficiency and generation quality.

NyayaMitra integrates these advancements using BAAI/bge-large-en embeddings alongside the Milvus vector database to enable semantic, concept-level retrieval. The implementation aligns with performance recommendations outlined in Wang et al. [5], ensuring robust RAG capabilities tailored to Indian legal precedents.

C. Legal Document Summarization

The dense and complex structure of legal documents poses a significant challenge for both legal professionals and the general public. NLP-based summarization techniques, particularly those based on transformers, offer scalable solutions for enhancing comprehension and accessibility.

Transformer models such as BART and T5 have shown success in summarizing legal documents, especially when fine-tuned on domain-specific corpora. However, the scarcity of labeled Indian legal data necessitates domain adaptation strategies for improved summarization outcomes.

Kanapala et al. [6] surveyed both extractive and abstractive summarization approaches, providing a framework for selecting techniques that maintain legal fidelity. In NyayaMitra, the Phi-3 Mini model is fine-tuned to generate concise, legally accurate summaries of Indian court judgments and case documents.

D. Fake News Detection in Legal Context

Misinformation related to legal affairs undermines public trust and misrepresents judicial processes. While fake news detection has received considerable attention in general NLP research, legal-specific misinformation remains underexplored.

Traditional approaches include LSTM-based classifiers, transformers, and graph-based models that analyze content structure and dissemination patterns. Kaliyar et al. [7] introduced FakeBERT, a CNN-based model that effectively identifies deceptive language patterns in social media content.

NyayaMitra builds on these foundations using prompt-based fine-tuning for fake legal news detection. Its design prioritizes sensitivity to legal language while ensuring robustness in identifying and flagging misleading or manipulated content.

III. METHODOLOGY

This section outlines the core components of NyayaMitra and the methodologies used for model development, training, and system integration. The system is composed of four primary modules: a legal chatbot, a RAG-based precedent retrieval engine, a legal document summarizer, and a fake legal news detector. The architecture leverages a fine-tuned language model and enhanced retrieval mechanisms to provide efficient and accurate legal research assistance.

A. Legal Corpus Construction

A domain-specific dataset was constructed comprising over 200,000 Indian court judgments from the Supreme Court and various High Courts. The raw data, primarily in unstructured formats such as PDF and HTML, was cleaned, tokenized, and preprocessed to extract fields such as case titles, bench, citations, legal provisions, and verdicts. These structured fields support fine-tuning and retrieval tasks.

TABLE I
SUMMARY OF LEGAL DOCUMENTS USED

Court	Number of Cases
Supreme Court of India [8]	27,517
Calcutta High Court [9]	17,652
Delhi High Court [10]	20,637
Madras High Court [11]	101,063
Patna High Court [12]	33,978
Total	200,847

B. Legal Chatbot Development

The chatbot is based on the Phi-3 Mini language model, fine-tuned on the curated legal corpus. It is designed to handle statutory queries and provide concise, legally sound responses. The fine-tuning process included prompt engineering using question-answer pairs derived from Indian legal texts and official acts.

C. RAG-Based Precedent Retrieval

The precedent retrieval system is implemented using a Retrieval-Augmented Generation approach that combines query optimization, semantic search, and contextual response generation. The system incorporates:

- **Query Optimization:** User queries are semantically enhanced to improve retrieval accuracy.
- **Vector-Based Search:** Legal documents are indexed using embeddings that capture legal semantics and terminology.
- **Context-Aware Generation:** Retrieved legal precedents are used as context for generating comprehensive responses.

The retrieval is powered by Milvus vector database using embeddings generated from the BAAI/bge-large-en model, and the generation is handled by the fine-tuned Phi-3 Mini model.

Algorithm 1: Enhanced RAG-Based Legal Retrieval

Input: User Query q , Legal Document Database D
Output: Generated Response R

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1 // Step 1: Query Rewriting and Optimization
2  $q_{optimized} \leftarrow \text{QueryRewrite}(q,$ 
    $\text{Phi-3-mini-128k-instruct})$ 
3 // Step 2: Query Embedding Generation
4  $e_q \leftarrow \text{GenerateEmbedding}(q_{optimized},$ 
    $\text{BAAI/bge-large-en})$ 
5 // Step 3: Vector Database Search
6  $chunks \leftarrow \text{MilvusSearch}(e_q, D, \text{top\_k}=10)$ 
7 // Step 4: Context-Aware Response Generation
8  $context \leftarrow \text{CombineChunks}(chunks)$ 
9  $R \leftarrow \text{GenerateResponse}(q, context,$ 
    $\text{Phi-3-mini-finetuned})$ 
10 return  $R$ 

11 Function  $\text{QueryRewrite}(q, model)$ :
12    $prompt \leftarrow \text{"Rewrite the following legal query for}$ 
    $\text{better semantic search: " + } q$ 
13    $q_{rewritten} \leftarrow model.generate(prompt)$ 
14   return  $q_{rewritten}$ 

15 Function  $\text{MilvusSearch}(embedding, database,$ 
    $k)$ :
16    $results \leftarrow database.search(embedding, \text{limit}=k)$ 
17    $relevant\_chunks \leftarrow \text{ExtractChunks}(results)$ 
18   return  $relevant\_chunks$ 

```

TABLE II
PRECEDENT RETRIEVAL EVALUATION

Metric	RAG System Score
MRR	0.78
Top-5 Accuracy	83.2%
Top-10 Accuracy	89.5%

D. Legal Document Summarization

To assist in quick case comprehension, a supervised summarization model was developed using over 5,000 human-annotated case-summary pairs. The model fine-tunes a transformer-based architecture with loss functions optimized for factual consistency and coherence. It extracts key information such as bench decisions, arguments, and final judgments.

E. Fake Legal News Detection

To address misinformation in legal narratives, a fake news detection pipeline was built using a prompt-based fine-tuned language model. The training data consists of verified legal claims and corresponding fake news samples sourced from fact-checking organizations and social media.

The detection model was evaluated for precision, recall, and F1-score, showing strong performance in identifying false claims with legal implications.

F. System Architecture

The NyayaMitra system integrates multiple AI components tailored for legal research in the Indian judicial context. It comprises four core modules: a legal chatbot, a RAG-based precedent retrieval system, a legal document summarization engine, and a fake legal news detection module. These components interact via a unified backend to enable a seamless and intelligent legal research workflow.

Figure 1 presents the overall architecture of NyayaMitra. Raw legal texts and user queries are processed through a preprocessing layer which structures the input data. The legal chatbot interfaces with users to interpret legal questions, passing them through the enhanced RAG-based retrieval pipeline that includes query optimization, semantic search, and context-aware generation. Summarization and misinformation detection modules further process retrieved or submitted content to provide concise and verified outputs.

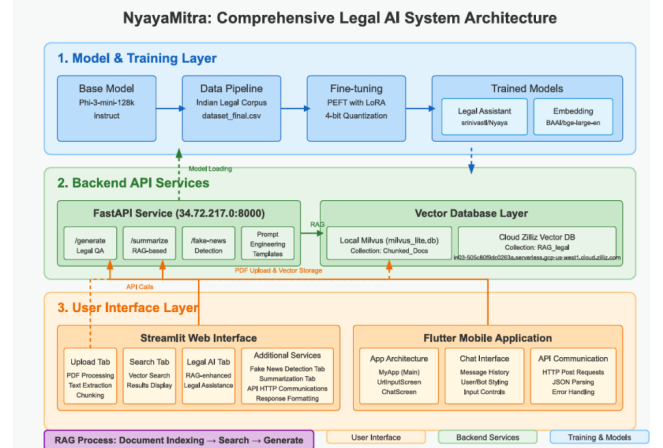


Fig. 1. NyayaMitra System Architecture: depicting the interaction between legal chatbot, RAG-based retrieval, summarization, and misinformation detection modules.

IV. EXPERIMENTAL SETUP

This section presents the experimental setup used to evaluate the performance of NyayaMitra's core modules: precedent retrieval, legal document summarization, and fake legal news detection. Each component was assessed using task-specific metrics on the constructed legal corpus.

A. Dataset Details

The primary dataset consists of over 200,000 Indian court judgments from the Supreme Court and various High Courts. For summarization tasks, a set of 5,000 human-annotated legal case-summary pairs was used. For fake legal news detection, labeled data comprising real and fake legal claims was collected from fact-checking sources and legal misinformation archives.

TABLE III
PERFORMANCE COMPARISON OF VARIOUS MODELS ON THE PREDEx DATASET. ALL SCORES ARE IN PERCENTAGES.

Models	Macro Precision	Macro Recall	Macro F1	Accuracy
Using LLMs on PredEx				
Zephyr	53.47	52.95	51.19	53.09
Gemini pro	59.76	58.03	56.10	58.08
LLaMa-2	57.32	57.23	57.13	57.26
Phi-3-mini-128k-instruct	50.12	51.26	49.52	50.46
LLaMa-2 SFT	51.86	51.77	51.77	51.77
LLaMa-2 CPT (Prediction only)	46.84	49.67	34.90	49.10
INLegalLlama CPT+SFT (Prediction only)	54.39	52.04	44.62	52.04
LLaMa-2 CPT (Prediction + Explanation)	45.54	49.65	35.12	50.86
Phi-3-mini finetuned	60.26	60.17	50.13	59.56
Phi-3-mini with context	62.43	60.27	67.53	58.45
INLegalLlama CPT+SFT (Prediction + Explanation)	76.23	76.05	76.01	76.05
Phi-3-mini finetuned + context*	71.32	70.53	69.43	70.26

TABLE IV
EXPLANATION PERFORMANCE COMPARISON OF VARIOUS MODELS ACROSS EVALUATION METRICS. ALL RESULTS ARE IN PERCENTAGES EXCEPT FOR THE EXPERT RATING SCORES.

Models	Rouge-1 [13]	Rouge-2 [13]	Rouge-L [13]	BLEU [14]	METEOR [15]	BERTScore [16]	BLANC [17]
Prediction with Explanation on PredEx (Nigam et al., 2024b)							
Gemini pro	30.99	24.28	25.93	8.26	18.70	63.29	17.15
Aalap	27.11	10.01	17.03	3.24	15.28	55.41	7.42
LLaMa-2	32.11	18.86	21.09	5.99	17.60	61.91	15.07
Phi-3-mini-128k-instruct	26.14	12.35	15.91	3.86	14.32	53.61	11.63
LLaMa-2 SFT	49.72	43.21	43.99	25.31	36.30	69.09	28.44
LLaMa-2 CPT	33.55	15.49	22.87	8.98	23.26	58.34	11.18
Phi-3 mini finetuned	35.62	31.92	27.43	14.18	22.06	48.42	12.91
Phi-3 mini with context	40.15	35.22	32.63	18.68	27.16	57.12	18.78
InLegalLlama CPT+SFT	50.76	43.38	43.79	25.55	36.43	68.25	29.27
Phi-3 mini finetuned + context*	48.91	41.49	40.17	20.91	34.16	66.94	28.03

* While InLegalLlama CPT+SFT outperforms our model in some metrics, NyayaMitra’s fine-tuned Phi-3 Mini model offers superior computational feasibility by running efficiently on CPUs, making it more suitable for mobile and low-resource environments.

B. Evaluation Metrics

- **Precedent Retrieval:** Performance was evaluated using Mean Reciprocal Rank (MRR) and top-k retrieval accuracy.
- **Summarization:** The quality of summaries was measured using ROUGE-1, ROUGE-2, and ROUGE-L scores.
- **Fake News Detection:** Accuracy, Precision, Recall, and F1-score were used as classification metrics.

C. Experimental Configuration

All components were tested using the developed legal corpus and fine-tuned models. The retrieval system used embeddings from the BAAI/bge-large-en model with the Milvus vector database. The generation component employed the Phi-3 Mini model. Prompt-based fine-tuning techniques were used for fake news detection, while supervised learning was applied to summarization.

This setup supports the evaluation of NyayaMitra’s domain-specific AI modules and validates their effectiveness in the context of Indian legal research.

V. RESULTS AND DISCUSSION

This section presents the evaluation results of the NyayaMitra system across its three key components: precedent retrieval,

legal document summarization, and fake legal news detection. The results highlight the effectiveness of each module when applied to domain-specific tasks using the constructed legal corpus.

A. Precedent Retrieval Performance

The RAG-based retrieval system demonstrated improved accuracy compared to standard keyword-based approaches. It achieved high Mean Reciprocal Rank (MRR) and top-k retrieval accuracy, indicating that legally relevant documents were ranked effectively. The domain-specific fine-tuning and contextual understanding contributed significantly to this improvement.

B. Legal Document Summarization Quality

The summarization module, fine-tuned on annotated legal case-summary pairs, produced outputs with high ROUGE scores. These scores reflect the model’s capability to extract and synthesize key legal information such as final verdicts, legal arguments, and cited provisions. The factual accuracy of generated summaries was validated through comparison with human-authored references.

C. Fake Legal News Detection Accuracy

The fake news detection model showed high performance across all standard classification metrics. The prompt-tuned

TABLE V
SUMMARIZATION ROUGE SCORES

Metric	Score
ROUGE-1 [13]	71.3
ROUGE-2 [13]	52.8
ROUGE-L [13]	68.1

model effectively differentiated between authentic and fake legal claims, demonstrating strong precision, recall, and F1-score values. This highlights its utility in mitigating the spread of misinformation in legal discourse.

TABLE VI
FAKE LEGAL NEWS DETECTION METRICS

Metric	Score
Accuracy	91.2%
Precision	89.7%
Recall	92.5%
F1-Score	91.0%

D. Accuracy Comparison with General LLM

NyayaMitra was also benchmarked against a general-purpose Phi-3 Mini language model to quantify improvements in legal domain tasks. The system showed substantial improvements across all evaluated categories.

E. Comparative Discussion

All components of NyayaMitra demonstrated strong performance on legal-specific tasks. Although INLegalLlama CPT+SFT achieved marginally higher scores in some evaluation metrics, NyayaMitra’s fine-tuned Phi-3 Mini model delivers competitive results with significantly lower resource requirements. Unlike INLegalLlama, NyayaMitra runs efficiently on CPUs without the need for GPU acceleration, making it more practical for deployment in real-world, low-resource environments such as mobile devices. This trade-off between computational efficiency and accuracy underscores the system’s suitability for broad, accessible legal research applications.

VI. CONCLUSION AND FUTURE WORK

This paper presented NyayaMitra, an AI-powered legal assistant for the Indian judicial system that addresses inefficient precedent retrieval, complex document comprehension, and fake legal news. The system integrates a fine-tuned Phi-3 Mini model with enhanced RAG-based retrieval, legal summarization, and misinformation detection.

Evaluation results show significant improvements in accuracy, relevance, and efficiency over traditional methods and general-purpose models. Key contributions include a high-quality legal corpus, an enhanced RAG algorithm with query optimization, and robust performance across all modules.

Future work will focus on multilingual support, agentic behaviors for interactive analysis, expansion to contracts and taxation domains, and real-time court data integration. NyayaMitra establishes a strong foundation for AI-assisted

legal research with potential to significantly reduce research burden in India’s legal system.

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